

q-q plot example

Data:

X_1	X_2	...	X_{10}
93	77	67, 72, 52, 83, 66, 84, 59	63

Ordered data:

Y_1	Y_2	...	Y_{10}
52	59	63, 66, 67, 72, 77, 83, 84	93

($Y_1, \dots, Y_{10}$ - order statistics)

Q-q plot : $(Y_r, z_{1-\frac{r}{n+1}})$ for $r=1, \dots, n$

ordered data from the table $\bar{V}_b$

For simplicity, let's compute for odd r 's only:

$r=1$

$(52, z_{1-\frac{1}{11}}) = (52, z_{0.91}) = (52, -z_{0.09}) = (52, \underline{-1.3})$

from the table

$r=3$

$(63, z_{1-\frac{3}{11}}) = (63, z_{0.73}) = (63, -z_{0.27}) = (63, \underline{-0.61})$

$r=5$

$(67, z_{1-\frac{5}{11}}) = (67, z_{0.55}) = (67, -z_{0.45}) = (67, \underline{-0.12})$

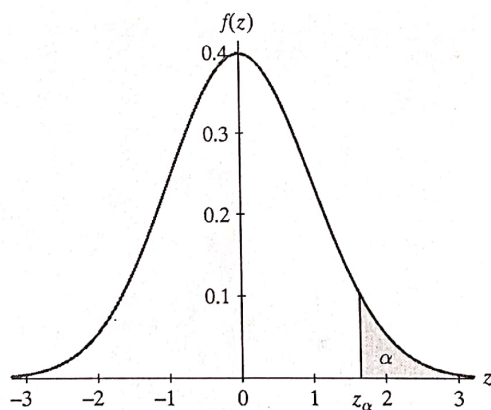
$r=7$

$(77, z_{1-\frac{7}{11}}) = (77, z_{0.36}) = (77, \underline{0.36})$

$r=9$

$(84, z_{1-\frac{9}{11}}) = (84, z_{0.18}) = (84, \underline{0.92})$

Table Vb The Standard Normal Right-Tail Probabilities



$P(Z > z_\alpha) = \alpha$

$P(Z > z) = 1 - \Phi(z) = \Phi(-z)$

z_α	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294

$r=5 \rightarrow$

$r=7 \rightarrow$

$r=3 \rightarrow$

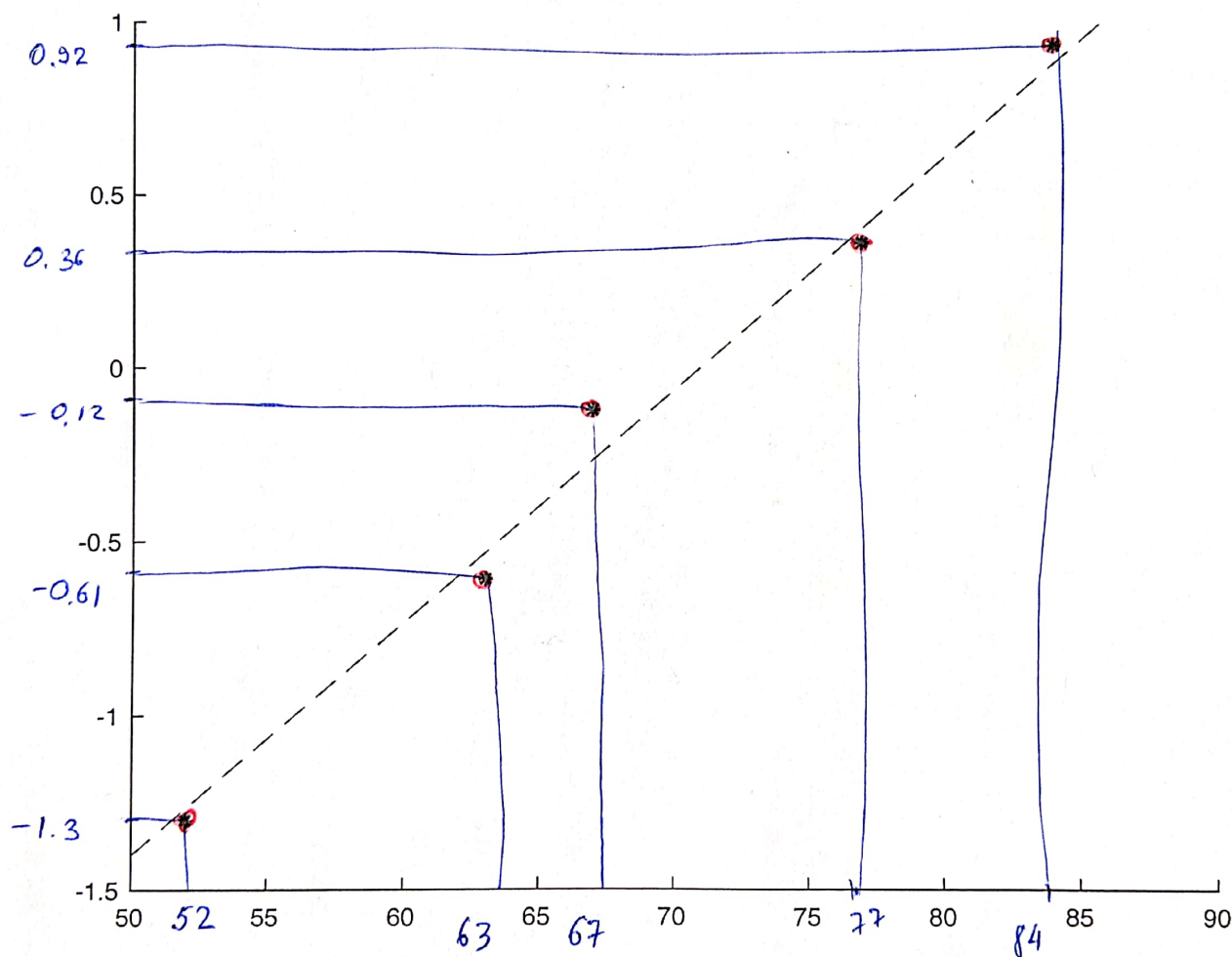
$r=9 \rightarrow$

$r=1 \rightarrow$

Red points are close to a straight line $\Rightarrow$

①

we can claim that data has approximately normal distribution



②

$$\frac{1}{\text{Slope}} \approx S$$

↑
sample standard deviation

$S > 0$, so ~~slope~~ slope > 0
 $\Rightarrow$ line is always $\nearrow$

Indeed, $\text{std}([52, 59, 63, 66, 72, 77, 83, 84, 93]) = \underline{13.3}$

$$\frac{1}{\text{slope}} \approx \frac{77-52}{0.36-(-1.3)} = \frac{25}{1.66} \approx \underline{15}$$